

INTRODUCTION TO CYBERSECURITY

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INTRODUCTION

Cyber security refers to the body of technologies, processes, and practices designed to protect networks, devices, programs, and data from attack, damage, or unauthorized access.

Cybersecurity is relevant to **Individuals, Organizations** and **Governments**.

CIA TRIAD

The CIA triad is the foundation for security, It is the same everywhere, as long as security is involved. They are:

Confidentiality: This means that data is kept private and secret. It means that only authorized or permitted persons have access to resources. Confidentiality is ensured by ***encryption***.

CIA TRIAD

Integrity: This means that data is not tampered with, whether it is at rest or in transit. It indicates that data is not changed at any point, whether it is sitting on a drive, or sent over a network. Integrity is checked using ***hashing*** or ***checksums***.

CIA TRIAD

Availability: Availability means that resources can be accessed by authorized users any time that they need it. Availability is ensured using ***access control***.

COMMON CYBERSECURITY THREATS

Viruses: a virus spreads within a device or a network by attaching itself to a host file. It needs to be interacted with before it can spread. An example of a virus is the Michelangelo virus of 1992.

Worms: worms spread without user interaction. They can spread within devices or networks on their own. An example of a worm is Conficker worm of 2008.

COMMON CYBERSECURITY THREATS

Ransomware: a ransomware takes data captive with a promise of restoring it when a *ransom* is paid. An example of ransomware is WannaCry of 2017.

Trojan: A trojan is a malicious software that disguises as a legitimate software. It does its work, but also carries out malicious activity. It could be a word processor that also acts as a keylogger.

COMMON CYBERSECURITY THREATS – PHISHING

Phishing: Deceptive messages designed to trick you into revealing personal info. It is common in emails, SMS (smishing), fake websites, voice calls (vishing) and social media DMs.

Signs of phishing:

- Urgent or threatening language (“Your account will be closed!”).
- Suspicious links (hover to check before clicking).
- Poor spelling, odd grammar, strange email addresses.

SPAM EMAIL

SPOT THE DIFFERENCE

there are 6 differences between the fake and real one, can you spot them?

FAKE

From: support@microsoft.co.uk
Sent: 16/01/2023 11:44
To: **Bob Smith** <Bob.Smith@company.com>
Subject: Urgent Action Needed!



Microsoft Account

Verify your account

We detected some unusual activity about a recent sign in for your Microsoft account. you might be signing in from a new location app or device.

To help keep your account safe. We've blocked access to your inbox , contacts list and calander for that sign in. Please review your recent activity and we'll help you secure your account. To regain access you'll need to confirm that the recent activity was yours.

[http://account.liive.com/ResetPassword.aspx](http://account.live.com/ResetPassword.aspx)

Thanks,
The Microsoft Team

REAL

From: support@microsoft.co.uk
Sent: 16/01/2023 11:44
To: **Bob Smith** <Bob.Smith@company.com>
Subject: Unusual Sign In Activity



Microsoft Account

Verify your account

We detected some unusual activity about a recent sign in for your Microsoft account bo*****@company.com. you might be signing in from a new location app or device.

To help keep your account safe. We've blocked access to your inbox, contacts list and calendar for that sign in. Please review your recent activity and we'll help you secure your account. To regain access you'll need to confirm that the recent activity was yours.

[Review recent activity](#)

Thanks,
The Microsoft Team

COMMON CYBERSECURITY THREATS – SOCIAL ENGINEERING

Cybercriminals exploit human psychology rather than technology. Manipulating people into breaking security rules.

Tactics include:

Pretexting – pretending to be an authority (bank officer, IT support).

Baiting – offering free downloads, infected USB drives.

Tailgating – following someone into a secure area.

COMMON CYBERSECURITY THREATS – PASSWORD ATTACKS

Hackers try to crack or steal login details.

Methods:

Brute Force: Guessing all combinations.

Credential Stuffing: Using leaked passwords from breaches.

Keyloggers: Recording every keystroke.

Weak Password Example: 123456, password, qwerty, iloveyou.

See *rockyou.txt* for leaked passwords.

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PASSWORD ATTACKS

To check for breached passwords or emails, you can use **HaveIBeenPwned**.

<http://haveibeenpwned.com/Passwords>

To check how strong your password is, you could use **Bitwarden Password Strength Tester**.

<https://bitwarden.com/password-strength/>

DoS/DDoS Attacks

Flooding a system with traffic so it becomes slow or crashes.

Often used to take down websites, apps, or company servers.

DDoS uses botnets (robots on a network) to attack servers.

Insider Threats

Threats from people within an organisation, employees, contractors or trusted insiders.

Types:

Malicious insiders: Employees who intentionally steal or leak data.

Accidental insiders: People who click phishing links or mishandle data.

CYBERSECURITY HYGIENE

- Use strong passwords (a mix of letters, numbers, and symbols).
- Two-Factor Authentication (2FA)
- Regularly update software
- Don't click unknown links or download suspicious attachments.
- Regularly back up data to external drives or cloud storage.